Abdullah Gül University

Electrical Electronics Engineering

SENIOR DESIGN PROJECT PROPOSAL FROM

Title of the Project	Enhancing Robot Navigation with Vision Language Models
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Date	08.03.2026

1. ABSTRACT AND KEY WORDS: The abstract should be 150-300 words.

While traditional mobile robot navigation systems excel at object detection and mapping using sensors such as LIDAR and YOLO, they lack the contextual common sense and reasoning capabilities required in complex and dynamic social environments. This project aims to enhance navigation capabilities by integrating Vision Language Models (VLMs) into a mobile robot's perception-decision pipeline. By leveraging the descriptive and reasoning power of VLMs, the goal is for the robot to not only detect obstacles but also interpret semantic details such as human intentions and social context. Within the scope of this study, real-time performance analysis of these models will be conducted on edge hardware like the Jetson Nano and cloud-based API solutions. The system will be tested in ROS and Gazebo simulation environments and compared with traditional SLAM and object detection-based methods. Ultimately, this study presents a more transparent and safe social navigation architecture where the robot can explain its decisions in natural language.

Key words: Vision Language Models (VLM), Mobile Robot Navigation, ROS, Edge Computing.

2. PURPOSE: The purpose of the project and the expected outcomes should be included in this section.

The primary purpose of this capstone project is to **evaluate and enhance the navigation capabilities of a mobile robot by integrating a Vision Language Model (VLM) into its perception-decision pipeline.** While traditional sensors like LIDAR and standard object detectors (YOLO/SSD) excel at spatial mapping and basic labeling, they often lack the contextual "common sense" required for complex decision-making in dynamic environments. This project aims to bridge that gap by leveraging the descriptive and reasoning power of VLMs.

Key objectives include:

- **Advanced Semantic Perception:** Investigating how VLMs can extract nuanced environmental data such as human intent, social context, or complex obstacle descriptions that traditional sensors might miss.
- **Decision-Layer Optimization:** Determining how natural language descriptions from the perception module can be translated into actionable navigation commands, leading to more "intelligent" and adaptive movement.
- **Performance Benchmarking:** Comparing the VLM-enhanced navigation system against traditional SLAM and object-detection-based methods to measure improvements in robustness, safety, and efficiency.
- **Feasibility Analysis:** Assessing the real-time processing capabilities of VLMs on onboard companion computers to ensure the robot can react promptly to environmental changes.

Ultimately, this project seeks to demonstrate that the synergy between vision and language can move mobile robotics closer to human-like environmental understanding.

3. SUBJECT: Please give a brief summary about the subject, including previous efforts. A few references are expected in this section.

Today, robots play critical roles in both social and industrial environments. In daily life, they appear as intelligent assistants capable of interpreting human intentions, analyzing emotions, and performing social navigation by understanding natural language commands. In industry, thanks to their visual perception and reasoning abilities, they increase production efficiency by carrying out advanced control tasks such as object recognition in complex assembly lines, precision manipulation, and operational outcome prediction.

A mobile robot must be equipped with three main modules: Perception, decision, and actuation. With the perception module, robots are required to perceive the environment by their sensors such as camera, LIDAR, and distance sensors. The decision layer is expected to generate the future actions of the robot, considering the outputs of the perception module and the given task. The actuation module provides the inputs for the motion mechanisms that are required for meeting the decision module's commands. All three layers are inter-connected in a robot, and each layer's performance affects significantly the other layers' adaptability and robustness.

Traditional navigation modules of mobile robots usually rely on a set of LIDAR sensors, monocular/stereo cameras, depth sensors, and distance sensors. Until a decade ago, robots had used mainly LIDAR sensors and mono cameras for mapping the environment and localizing themselves in these maps, also referred to as simultaneous localization and mapping (SLAM). Although LIDAR sensors provide accurate and precise depth information of objects around the robot through their projected sizes and shapes, they give a little clue about the types and identities of them. In some applications requiring precise navigation, especially in dynamic environments, a robot should identify the types of objects for collision avoidance and efficient navigation. Conversely, a camera can give extensive information about its view, including the identity, type, and shape of the objects. With the recent advances in object detection algorithms such as YOLO and SSD, objects can be detected in real time on a companion onboard computer.

However, object detection methods can only identify objects in the field-of-view of the camera but do not use reasoning which might be needed in several navigation algorithms. For instance, object detection models can detect whether a person is present in an image, but a more detailed description is not provided by these models, such as whether the person is moving or in which direction. Recently, vision language models (VLM) have transformed the computer vision domain by providing meaningful and more detailed descriptions of the camera views. By using advanced deep learning layers, VLMs can answer user prompts quickly and provide useful information about the objects, people, and other entities in the image frame. This information is vital to a mobile robot as they can be directly employed in the decision layer.

Vision Language Models (VLMs) are multimodal architectures that integrate visual perception capabilities with natural language reasoning. By semantically aligning visual data, these models enable robots not just to label scenes but to interpret and understand them at a high level. Unlike traditional computer vision systems, VLMs possess the capacity for object recognition, spatial reasoning, and understanding social cues, such as a person's activity or intent to interact. In robotic applications, this capability allows for contextual environment analysis, sensing human intentions, and planning complex tasks through natural language commands. Today, advanced models like GPT-4o, LLaVA, and Phi-3 Vision bring robotic perception and decision-making processes closer to a more flexible, transparent, and human-like understanding of the surroundings

In this project, mobile robot navigation performance improvements will be investigated by adding VLM modules in the robot's perception layer. We ask the following question: *Can a mobile robot's navigation be improved with VLMs?* and *what kind of environmental information can be extracted from a camera image through VLMs?*

References:

Ahmad, S., Hafeez, M., Zaidi, S. A. R., & School of Electronic and Electrical Engineering, University of Leeds, UK. (2026). Vision-Language models on the edge for Real-Time robotic perception [Research-article]. *arXiv, 2601–14921*. https://arxiv.org/abs/2601.14921v1

The gaps can be extended if needed.

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Sasabuchi, K., Wake, N., Kanehira, A., Takamatsu, J., & Ikeuchi, K. (2025). Agreeing to interact in human-robot interaction using large language models and vision language models. *2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 931-938. https://doi.org/10.1109/RO-MAN63969.2025.11217646

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Zhou, W., Tao, M., Zhao, C., Guo, H., Dong, H., Tang, M., & Wang, J. (2025). PhysVLM: Enabling visual language models to understand robotic physical reachability. *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 6940-6949. https://doi.org/10.1109/CVPR52734.2025.00651

4. SOCIAL IMPACT: This section should include how your proposal addresses domestic and global issues.

The integration of Vision Language Models (VLMs) into robotic navigation is a critical step toward the social acceptance of autonomous systems in human-centered environments. While the inability of traditional navigation systems to communicate their decision-making processes often leads to user distrust, this project aims to provide real-time, human-readable justifications for navigation behaviors by leveraging the reasoning power of VLMs. As emphasized in literature, perceiving micro-social cues such as a person being engaged in a phone call and making context-aware decisions like waiting or rerouting is fundamental to socially aware navigation. To overcome the limited onboard processing capacity of robots, the proposed cloud-based architecture enables the analysis of complex multimodal streams by utilizing the high compute power of state of the art models with billions of parameters. This transparency and social awareness will ensure that robotic intentions are predictable for humans, thereby enhancing the social integration, safety, and adherence to social norms of these technologies on a global scale, particularly in healthcare and service sectors.

5. METHOD: Please explain the method that you will follow for the Project.

The methodology of this project is structured around integrating high-level semantic reasoning into a mobile robot's navigation stack, transitioning from virtual simulations to real-world hardware implementation. The core of the approach involves replacing or augmenting traditional object detection with Vision Language Models (VLMs) to achieve a more human-like environmental understanding.

The gaps can be extended if needed.

Initially, the project will leverage ROS (Robot Operating System) as the primary middleware to manage communication between the perception, decision, and actuation modules. To ensure safety and efficiency, initial testing will be conducted within Gazebo simulations. This virtual environment allows for the modeling of complex scenarios such as social navigation and dynamic obstacle avoidance where the VLM's ability to interpret human intent and social context can be evaluated against traditional SLAM methods.

The integration of VLMs will focus on transforming visual data into actionable intelligence through specialized prompt engineering. Instead of simple labels, specific prompts will be used to analyze image frames, asking questions like "Is the path blocked by a moving person?" or "Describe the nature of this obstacle". Because VLMs require significant computational power, the project will investigate a hybrid processing model. This involves utilizing the Intel Jetson Nano for local sensor fusion and basic tasks, while offloading complex reasoning to the Nvidia Cloud to ensure the robot can react promptly to environmental changes.

Finally, the methodology concludes with empirical testing on an experimental testbed using the Rosbot 2.0 platform. This phase focuses on Decision-Layer Optimization, where the natural language descriptions generated by the VLM are translated into specific navigation commands. By comparing these results with traditional object-detection (YOLO/SSD) benchmarks, the project will assess the feasibility of using vision-language synergy to enhance the robustness and safety of mobile robotics in everyday environments.

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6. SUCCESS CRITERIA:

The primary success criterion of the project is the ability of a mobile robot to generate meaningful and actionable information for navigation by utilizing camera frames supported by Vision Language Models (VLMs), going beyond traditional methods. The most critical indicator of the project's success will be the robot's ability to perceive its environment not merely as a geometric plane, but within a semantic context.

In this scope, the efficiency of the VLM integration within the robot's perception layer will be tested. The system is expected to interpret the raw data provided by traditional sensors (LIDAR, standard object detectors) to accurately identify the attributes and dynamic states of entities in the environment. The accuracy of the decision-making mechanisms exhibited by the robot in complex scenarios, the fluidity of its navigation, and its compliance with safety protocols will serve as the fundamental evidence of achieving the project's technical objectives.

Finally, the real-time operational capability and the stability of the proposed system on hardware will be evaluated. For the success of the project's application, it is essential that the data received from the VLM is converted into navigation commands without delay on edge units like the Jetson Nano or within cloud-based architectures. The demonstration of more adaptive and intelligent mobility through VLM support in both simulations and physical tests will certify the successful completion of the project.

7. EXTERNAL FUNDING OR SUPPORT Please indicate if you have any financial or advisory support from any external institution or company

Nothing.

The gaps can be extended if needed.

8. Timeline for the proposed work should be included in this section. Please have at least 3 work packages for each term.

TIMELINE (Months)

Work package name	Fall Term					Spring Term				
	9	10	11	12	01	02	03	04	05	06
Literature Review						X	X			
Setting up and configuring the VLM on a local development environment (Laptop)							X	X		
Testing perception and reasoning modules in Gazebo simulation							X	X		
Development of robot control and localization algorithms within ROS simulations							X	X		
Integration of VLM-based semantic data with robot control algorithms								X	X	
Transitioning and optimizing the software stack for the Jetson Nano board	X								X	X
Real-world performance testing and validation on mobile robot platforms		X	X	X						
Analysis of test results, repetition of low-performance tests with new methods, and writing a publishable conference paper.			X	X	X					

The gaps can be extended if needed.

9. BUDGET AND JUSTIFICATION:

9.1. Budget Table

Name/Definition	Quantity	Unit Price (TL)	Price Price (TL)
1- Intel Jetson Nano	2	360	720
2- Rosbot 2	3	750	2250
3- Mono camera (Nano camera)	1	150	150
4- Laptop	2	200	400
5- Cabling, soldering, etc.	2	-	-
6- Nvidia API (if paid version)	3	200	600
		Price including VAT	
		Overall Cost	???

(NOTE: All except item 6 are available in DIREC Lab.)

9.2. JUSTIFICATION:

Intel Jetson Nano: To serve as a high-performance edge computing node for real-time robotic perception and low-latency local inference by bringing computation closer to the data source.

Rosbot 2: To function as the primary embodied robotic testbed for validating perception, social navigation, and human-robot interaction pipelines in real-world scenarios.

Mono camera (Nano camera): To capture real-time visual observations and multimodal sensor streams required for scene interpretation, social cue recognition, and VLM-based reasoning.

Laptop: To serve as the central operator interface for rendering live video feeds, running simulations, and monitoring system-level performance metrics such as end-to-end latency and task accuracy.

Cabling, soldering, etc.: To establish reliable physical integration and data communication channels between onboard sensors, actuators, and the edge processing unit.

Nvidia API: To leverage cloud-based, large-scale VLM models for complex semantic reasoning and deep scene interpretation tasks that exceed the computational capacity of on-device edge hardware.

The gaps can be extended if needed.